MILITARY SPECIFICATION

BODY ASSEMBLY, TANK PERISCOPE: 10513599

1. SCOPE

1.1 This specification covers one type of periscope body assembly which, when assembled to Head Assembly 10513625, forms an infrared unity-power binocular periscope (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-E/1049	Electron Tube, Image Converter
MIL-O-13830	Optical Components for Fire Control Instruments, General Specification Governing the Manufacture, Assembly and Inspection of
MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel; Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-P-46320	Power Supply: 10516158 (Transistorized)
MIL-C-46369	Converter-Regulator: 10516120

STANDARDS

Military

MIL-STD-105	Sampling Procedure and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

FSC 6650

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DRAWINGS

U.S Army Munitions Command

F10513599	Body Assembly: 10513599
IEL-10513599	List of Inspection Documents for Body Assembly, Tank Periscope: 10513599

PACKAGING DATA SHEET

MIL-P-14232/P10513599	Packaging of Body Assembly: 10513599

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The body assembly shall be manufactured in accordance with Drawing F10513599 and drawings pertaining thereto.

3.2 General specification.- The contractor shall be responsible for adherence to, and compliance with, the requirements of MIL-F-13926 specified below:

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Electrical and electronic assemblies
(e) Part identification and marking
(f) Workmanship
(g) Soldering electrical connections

3.3 Image tubes.- The image converter electron tubes shall conform to MIL-E-1/1049.

3.4 Power supply.- The transistorized power supply shall conform to MIL-P-46320.

3.5 Converter.- The voltage converter-regulator shall conform to MIL-C-46369.

3.6 Environmental.

3.6.1 Storage temperatures.- The body assembly shall show no evidence of physical failure such as glass breakage, bonding failure, or casting failure due to storage at temperatures ranging from -80° to +155°F.

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3.6.2 Rigidity.- The following requirement shall be met during and subsequent to exposure to vibration at 30 ± 5 cycles per second and an amplitude of 0.0625 ± 0.0156 inch (0.125 ± 0.0312 inch total excursion). The catches shall be capable of securing the body assembly rigidly against the locating surfaces of the head and mount with which it is intended to be used.

3.6.3 Catches and switch (operability).- The two catches and the electrical switch knob shall be manually operable at temperatures ranging from -40° to +150°F.

3.7 Cleanliness.- The body assembly shall meet the cleanliness requirements of MIL-O-13830 and there shall be no more than three particles of dirt visible when viewing through the eyepiece. The area of each particle shall not be in excess of an area equivalent to a size ten dig. There shall be no foreign matter within the body assembly which interferes with observation or adversely affects image quality.

3.8 Sealing.- The body assembly shall withstand an internal pressure of 5 ± 0.10 pounds per square inch gage with no evidence of leakage.

3.9 Image steadiness.- This requirement shall be met with 18 to 30 Vdc applied to the converter and subsequent to allowing a 15 second time lapse to permit stabilization of the image tube. There shall be no visible flicker or movement of the image.

3.10 Collimation.- The requirements of 3.10.1 and 3.10.2 shall be met with the body assembly positioned as follows:

(a) The two retainers (Part No. 10513560) in the top position.
(b) The machined top surfaces of the four pads in a horizontal plane.

3.10.1 Dipvergence.- With a line of sight incident to each objective and perpendicular to a horizontal plane, the angle formed by the two emergent lines of sight shall not exceed 15 minutes of arc when measured horizontally.

3.10.2 Divergence.- With a line of sight incident to each objective and perpendicular to a horizontal plane, the two lines of sight shall diverge within the limits of 5 and 30 minutes of arc.

3.11 Image tilt.- The image of a target line originally perpendicular to a plane, defined by the 3/4" x 3/4" with 3/8" radius surface adjacent to Valve Assembly 8201751, shall emerge perpendicular to the same plane within 45 minutes of arc.

3.12 Eyepiece focus.- Eyepiece focus shall be set between the limits of -0.50 and -1.0 diopter.

3.13 Resolution.- Resolution on each optical axis shall be 7.0 minutes of arc or less.

3.14 Interior atmosphere.- The interior atmosphere of the accepted body assembly shall contain nitrogen having a dew point at least as low as -25°F.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. Inspection of components and sub-assemblies not covered by specifications or Supplementary Quality Assurance Provisions (SQAP) shall be in accordance with 4.2.2. Definitions of inspection terms not otherwise defined herein shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Inspection lot.- Unless otherwise specified by the contracting officer, inspection lot size, formation, and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Examination and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926 and the Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents, IEL-10513599. Examinations and tests related to Section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with MIL-STD-105, and the inspection level and sampling plans specified in Table I titled "Classification of Defects". Examinations and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

TABLE I - CLASSIFICATION OF DEFECTS

Use Inspection Level II of Table I with
Sampling Plan Table II-A of MIL-STD-105

		REQUIREMENTS	TEST PROCEDURES
CRITICAL: NONE			
MAJOR: AQL 1.0%			
101.	Rigidity (vibration)	3.6.2	4.5.2.2
102.	Cleanliness	3.7	4.5.3.2
103.	Image steadiness	3.9	4.5.3.1.1

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		REQUIREMENTS	TEST PROCEDURES
104.	Dipvergence (collimation)	3.10 & 3.10.1	4.5.3.3
105.	Divergence (collimation)	3.10 & 3.10.2	4.5.3.4
106.	Image tilt	3.11	4.5.3.5
107.	Eyepiece focus	3.12	4.5.3.6
108.	Resolution	3.13	4.5.3.7

MINOR: NONE

4.2.3 Disposition of non-conforming product.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.3 Control tests.

4.3.1 General.- One body assembly shall be selected at random as a control sample from each 100 produced, or from each month's production, whichever occurs first. The control sample shall have successfully met all other acceptance tests specified herein, and shall then meet the following requirements and tests:

CONTROL TEST		REQUIREMENT	TEST PROCEDURES
301.	Fabrication	3.1	4.5.1
302.	General specifications	3.2	MIL-O-13830 & MIL-F-13926

4.3.2 Environmental and functional.- Three body assemblies shall be selected at random as a control sample from each 50 produced, or from each month's production, whichever occurs first. The control sample shall have successfully met all other acceptance tests specified herein, and shall then meet the following requirements and tests.

CONTROL TEST		REQUIREMENT	TEST PROCEDURES
303.	Storage temperature	3.6.1	4.5.2.1
304.	Catches & switch (operability)	3.6.3	4.5.2.1
305.	Sealing	3.8	4.5.2.3
306.	Interior atmosphere	3.14	4.5.4

4.3.3 Control test failure.- Should any one item of a control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

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4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall maintain all required measuring and testing equipment in accordance with the applicable requirements specified in MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.4.2 Contractor furnished inspection equipment.

4.4.2.1 Government design.- All inspection equipment required by drawings forming a part of the contract and not provided by the Government shall be supplied by the contractor in accordance with the requirements of 4.4 and the List of Inspection Documents IEL-10513599.

4.4.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" requirements of MIL-F-13926. Concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit, and will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment.

4.4.2.3 Devices used for environmental testing.

4.4.2.3.1 Extreme temperature and rigidity.- The testing devices used to determine conformance with the extreme temperature and rigidity requirements of 3.6.1, 3.6.2 and 3.6.3 shall conform to the "Test facilities" requirements outlined in MIL-F-13926.

4.4.2.3.2 Sealing.- The device used to perform the sealing test (pressure test) shall conform to the accuracy requirements of 3.8, and as depicted on Drawing F7680673.

4.4.2.4 Devices to test optical performance requirements.- The device used to determine the acceptability of the optical requirements shall provide for the positioning of the four mounting pads in a plane perpendicular to the incident line of sight. This device shall also provide two target images, each consisting of a horizontal and vertical line and appropriate resolving target pattern for resolution testing (see MIL-O-13830). The intersecting point of these lines shall be placed on

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the optical axis of a projector collimator. The optical axis of each collimator projector shall lie in the same plane when both vertical lines are parallel to a plumb line. Illumination of these targets (reticles) shall be by means of a Tungsten radiation source and an infrared filter. This filter shall not pass visible light rays below 0.65 microns. This testing device shall also incorporate features for holding viewing collimators (double policeman). The reticles of the viewing collimators shall consist of a horizontal and vertical center line, the intersecting point of which shall be on the optical axis. The reticles shall also be provided with a grid pattern located on the optical axis. Each horizontal and vertical line shall represent an angle of five minutes. The optical axis of the viewing collimators shall be positioned perpendicular to the optical axis of the projector collimators when the vertical reticle lines are parallel to a plumb line. This device shall also provide a power supply of 24 Vdc to illuminate the body assembly under test and provide a power source to illuminate the reticles of the two projector collimators.

4.5 Test methods and procedures.

4.5.1 Fabrication.- The body assembly shall be subjected to a visual, tactile and dimensional inspection to determine compliance with the applicable drawings and requirements of this specification. Dimensional inspection shall be performed using standard measuring equipment.

4.5.1.1 Inspection of assemblies and components.- Assemblies and components covered below shall be subjected to 100 percent visual and tactile examination to determine evidence of defects or damage resulting from shipping or storage handling prior to final assembly. This type of examination shall be in addition to the respective examination specified below.

4.5.1.2 Image tube.- The image tube shall be visually examined for evidence that it has met the requirements of MIL-E-1/1049 as specified in 3.3.

4.5.1.3 Power supply.- The power supply shall be visually examined for evidence that it has met the requirements of MIL-P-46320 as specified in 3.4.

4.5.1.4 Converter.- The converter shall be visually examined for evidence that it has met the requirements of MIL-C-46369 as specified in 3.5.

4.5.2 Environmental tests.

4.5.2.1 Storage and operability temperatures.- This test shall be conducted in accordance with Procedure I of the temperature cycling test in MIL-F-13926 at the temperatures specified in 3.6.1 and 3.6.3. When performing the operability test of the clamps, the body assembly shall be mounted and secured on a device similar to the device shown in Figure #1. While exposed to the temperature extremes specified in 3.6.3 both catches and the electrical switch knob shall be manually operated to determine compliance with the requirements of 3.6.3. Upon completion of the specified tests and return to standard ambient temperature, (60° to 90°F) the body assembly shall be inspected for evidence of physical damage as specified in 3.6.1.

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4.5.2.2 Rigidity.- This test shall be accomplished by means of a holding device as shown in Figure 1, clamped to a vibration table. Clamp the body assembly in the holding device by means of the catches. The body assembly shall be vibrated in a vertical plane at the amplitude and frequency specified in 3.6.2 for a period of not less than 5 minutes. The vibration amplitude shall be measured at the extreme bottom portion of the body assembly as positioned in the fixture. During vibration, check rigidity of the body assembly in the holding device using a stroboscope to produce the optical effect of stopping the vibration motion. The body assembly shall remain secure against the holding device in conformance with the requirements of 3.6.2. Subsequent to vibration the body assembly shall be subjected to a visual and tactile examination and shall show no evidence of physical damage or failure.

4.5.2.3 Sealing.- The body assembly shall be pressurized to the psig value for a period of not less than 5 minutes. This pressure test is performed using a pressure tester, the accuracy of which shall be within the limits depicted on Drawing F7680673. The pressure test shall be performed using set-up and operation instructions specified on the fixture drawing to determine compliance with 3.8.

4.5.3 Performance tests.

4.5.3.1 Orientation.- Position the periscope body in a holding device as defined in 4.4.2.4 to insure conformance with the orientation specified in 3.10(a), (b) and 3.11.

4.5.3.1.1 Image steadiness.- Image steadiness shall be inspected for acceptance using a 1.5 Vdc power source in conjunction with the power supply providing 28 Vdc as specified on Drawing F10513599 with the converter. The actual test is performed by energizing the projector collimator's reticle. Insert a 1.5 Vdc dry cell (C type) battery into the body assembly to energize the infrared tube. Place the double-pole double-throw switch to 1.5-volt (ON) position. Allow the time specified in 3.9 for stabilization of the target image. There shall be no visible flicker or movement of the target image as specified in 3.9. At the completion of the test, place the double-pole double-throw switch to the OFF position, remove the 1.5-volt battery and insert the converter into the body assembly. Apply 24 Vdc to the 24-volt receptical and place the double-pole double-throw switch in the 24 Vdc (ON) position. Again observe the target image after allowing the time specified in 3.9 for stabilization of the image tube. There shall be no visible flicker or movement of the target image as specified in 3.9.

4.5.3.2 Cleanliness.- The body assembly shall be inspected for the cleanliness requirements of 3.7 while positioned in the testing device outlined in 4.4.2.4 with the 24 volt power source applied. Again allow the tube to stabilize itself as required in 3.9 and remove the viewing collimators. Observe cleanliness by sighting through the eyepiece assemblies. Irregularities in the fluorescent screen of the tube will be in the form of hot and cold spots and will appear as light and dark spots. These are not to be considered as dirt. Tube irregularities will show as varying light and dark spots whereas dirt on the optical surfaces will always remain dark. Cleanliness shall be in conformance with 3.7 when determined by the above procedure.

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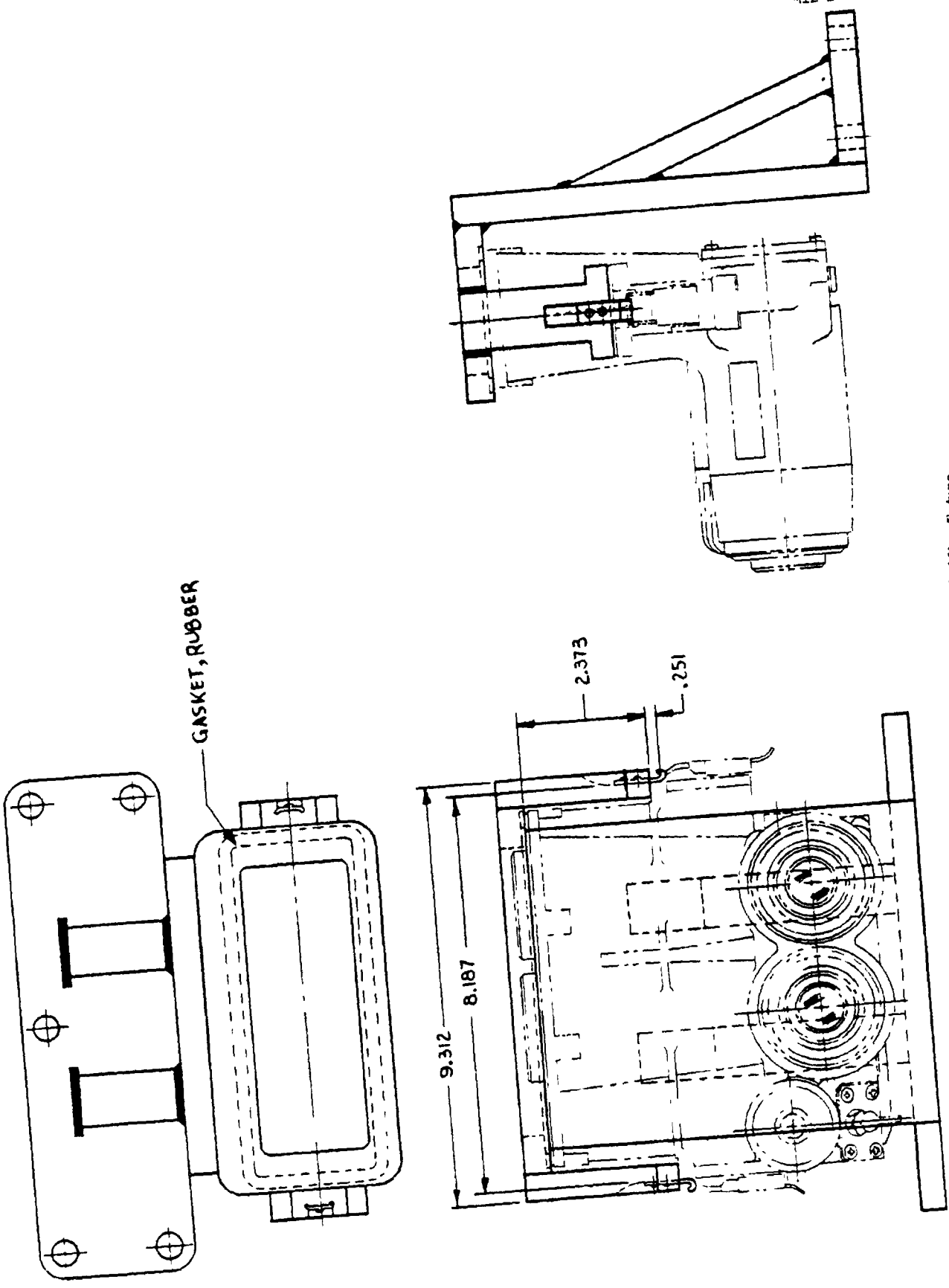


Figure 1. Rigidity holding fixture.

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4.5.3.3 Dipvergence.- With the body assembly secured in a testing device and positioned as specified in 4.5.3.1 dipvergence shall be measured by observing coincidence between the intersecting point of the projected target and the intersecting point of the horizontal and vertical reticle line on the optical axes of the viewing collimators. Coincidence between the intersecting point passing through each optical system shall be in the same horizontal plane as the horizontal reticle line of the viewing telescope within the limits outlined in 3.10.1, as measured by the graduated reticle of the viewing collimators.

4.5.3.4 Divergence.- This test shall be performed using the set-up specified in 4.5.3.3, however, coincidence must be observed between the intersecting point of the projected target and the intersection of the horizontal and vertical centerline of the viewing collimator. Coincidence between the intersecting point passing through each optical system shall not diverge by more than the limits specified in 3.10.2 as measured on the graduated reticle of the viewing collimators.

4.5.3.5 Image tilt.- This test shall be performed using the set-up specified in 4.4.2.4. Image tilt shall be measured by observing coincidence between either the horizontal or vertical reticle line of the projected targets and the horizontal or vertical reticle line of the viewing collimators. Image tilt shall not exceed the limit specified in 3.11 as measured on the graduated reticle of the viewing collimators.

4.5.3.6 Eyepiece focus.- This test shall be conducted with the aid of a calibrated dioptometer. The eyepiece of the dioptometer shall be adjusted for best focus on the dioptometer reticle with the dioptometer set to zero and parallax removed for the target distance selected. Using the dioptometer, sight through each eyepiece of the body assembly. The target image in each eyepiece of the body assembly shall be in the best focus within the tolerance specified in 3.12 as measured on the dioptometer.

4.5.3.7 Resolution.- The test shall be performed by using the set-up outlined in 4.5.3.1. The resolution test is performed by loosening the body assembly and positioning each optical axis in coincidence with the intersecting point of the projected target lines. With the aid of a viewing telescope (dioptometer) observe the projected definition pattern. They must be resolved in each optical system in each of the four meridians within the limits outlined in 3.13.

4.5.4 Interior atmosphere.- Upon satisfactory completion of the preceeding tests, the body assembly shall be flushed with dry nitrogen at a pressure of between 0.1 and 0.25 psig using Pressure Test Fixture, F7680673. The flushing shall continue until the gas bled from the body assembly has a dew point at least as low as minus 25 degrees Fahrenheit in compliance with 3.14. The body assembly shall then be sealed.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P10513599. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The Tank Periscope Body Assembly: 10513599 is used together with Tank Periscope Head Assembly: 10513625 to form the XM48 Tank Periscope The XM48 Tank Periscope is primarily intended for use, in conjunction with two infrared headlights, as the driver's night viewing system in the XM551 Vehicle.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Selection of applicable levels of preservation, packaging and packing
(c) Applicable stock number
(d) Applicable packaging data sheet number (see 5.1)

Custodian:

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Preparing activity:

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Project No. 6650-A150Y

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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